

Feedback Memo — Step 7: Conversion-Channel Robustness Grid and Reframing

HDI-FN project · Closing the indicator-selection vulnerability · One 2×2 grid, one new subsection, one reframing

Why this step exists

The OECD "digital laggard" finding (Finland −4, Denmark −3, Germany −3) rests on a single indicator: hi-tech exports, which measures export composition, not citizen or state digital capability. Finland tops DESI and OECD digital-government tables, so a referee will call our finding an artifact of indicator selection. Separately, penetration enters the digital sub-index twice (directly, and inside $T_{\text{egov}}^{\text{adj}}$) — a double-counting objection. A single 2×2 robustness grid closes both: {4-input, 3-input without hi-tech} × { $T_{\text{egov}}^{\text{adj}}$, raw EGDl}.

What the trial run found (pending your verification)

Double-counting axis: ordinally irrelevant. T^{adj} vs raw-EGDI variants produce identical rankings (one adjacent Germany–Netherlands swap in the 3-input case). One sentence in §4.6 retires that objection. **Hi-tech axis: decisive.** Without hi-tech, Finland −4 → −1, Denmark −3 → +1, Germany −3 → −1. The trio are reach-and-provision leaders; their entire negative reordering flows through the conversion channel. South Korea's +6 splits into +3 (reach-and-provision) + 3 (conversion) — the flagship gain is broadly based. Iceland's 1st place is partly conversion-driven (4th without hi-tech; Norway leads the 3-input ordering). Bangladesh has a defined 3-input score: rank 17, identical to its standard HDI position. Spearman vs standard HDI: 0.921 (4-input) / 0.968 (3-input).

Your tasks

#	Task	Standard
1	Reproduce the grid independently from Table_1_Raw_Digital.csv and Table 4 HDI dimensions; check every rank against Robustness 5 Digital Variant Grid.csv (shared). Confirm the 4-input/ T^{adj} column reproduces published Table 4 exactly, and the four Spearman values.	Cell-by-cell; script committed to replication folder
2	LaTeX: new §4.6 "Sensitivity to the conversion channel." Report the grid; frame the 3-input variant as a decomposition device — each country's rank change = reach-and-provision component (3-input Δ) + conversion component (residual). Include the Iceland caveat (one sentence) and the Bangladesh supplementary rank.	Draft passages will be supplied; integrate and compile
3	Reframing sweep: abstract, §1, §3.4, §5.2, §6. Replace "digital laggard" language with digital conversion laggard ; the abstract's "Finland losing four" stays but gains the conversion qualifier. Promote the §5.2 sentence ("infrastructure investment does not automatically produce export sophistication") from gesture to stated finding, citing §4.6 as evidence. South Korea: add the +3/+3 decomposition to §5.2.	List every changed line in your report
4	Consistency check across Steps 5–7: §4 now has subsections 4.1–4.6 (full triple = 4.5, this grid = 4.6). Confirm numbering, cross-references, and that §2.8 announces all robustness checks it now contains.	Compiled PDF with no broken \ref

Watch-outs

Do not drop hi-tech from the main composite. The 4-input HDI-FN stays as the headline — our conceptual claim (Yazd et al.) is that economic conversion is constitutive of digital capability. The 3-input variant is a robustness and decomposition instrument, not a replacement. Bangladesh stays excluded from the headline ranking (the composite's definition requires all four inputs); its 3-input rank is reported in the supplement only. Make sure the abstract and §4.6 cannot be read against each other: if the abstract says Finland loses four ranks, it must say *through the conversion channel*. Norway leading the 3-input ordering goes in §4.6, not the abstract.

Sequencing

Step 7 is independent of the Step 5 OSI/TII verification and the Step 6 Φ work; all three can run in parallel and land in the same revision. Together they close the three main methodological exposures: the incomplete triple (5), the algebraic reduction of the facade ratio (6), and the single-indicator dependence of the OECD finding (7). After these, the methods section is referee-ready and we move to the smaller fixes list.